Notes: Antràs (QJE, 2003)
Firms, Contracts, and Trade Structure

Contents

1 Big picture

- **Question.** Why is such a large share of world trade conducted within firm boundaries, and why is intrafirm trade especially concentrated in capital-intensive industries and capital-abundant countries?
- **Core empirical facts.** U.S. intrafirm imports are higher in capital-intensive industries and higher from capital-abundant countries.
- **Core theoretical mechanism.** The paper combines incomplete contracts and property rights with a Helpman-Krugman trade model. Ownership gives residual control rights, and residual control rights matter because capital and labor investments are relationship-specific and noncontractible.
- **Ownership logic.** Capital investments are easier to share or control by the final-good producer than labor investments. Therefore, vertical integration mitigates the most important holdup distortion in capital-intensive production, while outsourcing is relatively better in labor-intensive production.
- **Trade logic.** Capital-abundant countries specialize relatively more in capital-intensive intermediates, and those intermediates are precisely the ones produced inside firm boundaries. Therefore capital-abundant exporters have higher intrafirm trade shares.
- **Empirical test.** The paper tests the model with U.S. related-party import data by industry and country. It finds strong positive relationships between intrafirm import shares and both industry capital intensity and country capital abundance.
- **Economic takeaway.** The boundaries of firms are not separable from trade patterns. Trade structure depends on the allocation of control rights inside multinational firms.

2 Incremental contribution of the paper

- **From trade costs to firm boundaries.** Before this paper, international trade theory could explain why firms locate production abroad, but generally treated firm boundaries as irrelevant. Antràs makes the multinational boundary decision endogenous.
- **Integration of two literatures.** The paper combines the Grossman-Hart-Moore property-rights theory of the firm with Helpman-Krugman monopolistic-competition trade theory.
- **Property-rights foundation for intrafirm trade.** The paper provides a microfoundation for why some international transactions occur inside multinational firms while others occur at arm's length.
- **New comparative advantage channel.** Classical factor endowments affect not only what countries trade, but also whether trade occurs inside or outside firm boundaries.
- **Empirical innovation.** The paper uses U.S. related-party trade data to document and test systematic patterns in intrafirm imports by both industry capital intensity and country capital-labor ratios.
- **Capital versus labor investment asymmetry.** The key new organizational assumption is that capital investments are more transferable/shareable than labor investments. This asymmetry turns capital intensity into a determinant of vertical integration.
- **Bridge to multinational-firm theory.** The paper opens a research program in which ownership, contracts, and global sourcing decisions are jointly determined.

Interpretation: The incremental contribution is not merely the documentation of intrafirm-trade correlations. It is the structural logic that links incomplete contracting, ownership, factor intensity, factor endowments, and observed trade flows in a single model.

3 Data and measurement

3.1 Intrafirm trade data

The paper uses data on U.S. intrafirm imports, defined as imports that occur between related parties. The central dependent variable is the share of intrafirm imports in total U.S. imports:

$$S_{i-f} = \frac{\text{related-party imports}}{\text{total imports}}.$$

The paper studies this share in two ways:

- by manufacturing industry, averaged over 1987, 1989, 1992, and 1994;
- by exporting country, using 1992 data.

Interpretation: This share is the empirical proxy for the organizational form of trade. A higher share means that more imports are internalized within multinational firms rather than transacted through arm's-length suppliers.

3.2 Industry capital intensity

Industry capital intensity is measured as the log of the ratio of capital stock to total employment in the exporting industry:

$$\ln(K/L)_m.$$

The paper uses U.S. industry data to proxy for technological factor intensity.

Interpretation: The theory predicts that capital-intensive industries should be more vertically integrated because integration gives the final-good producer stronger incentives to make transferable capital investments.

3.3 Country factor endowments

Country capital abundance is measured by the log of the exporting country's physical capital stock divided by its total number of workers:

$$\ln(K/L)_j.$$

The paper also controls for country size, human-capital abundance, corporate taxes, economic freedom, openness to FDI, and openness to trade.

Interpretation: Country factor endowments matter because capital-abundant countries specialize in capital-intensive intermediates, which are exactly the industries predicted to be integrated.

3.4 Motivating organizational evidence

Table I reports evidence from British affiliates of U.S. multinationals showing that parent firms exert much stronger influence over financial decisions than employment/personnel decisions. For example, parent influence is strong in dividend policy and royalty payments, but much weaker in wage increases, layoffs, and hiring.

Interpretation: This supports the modeling asymmetry: capital and financial decisions are more transferable and controllable by headquarters than labor and personnel decisions.

4 Model and theoretical specifications

4.1 Timing

The model has five dates. At t_0, the final-good producer chooses ownership, chooses who rents capital, and sets a transfer. At t_1, capital and labor investments are made. At t_2, the intermediate input is produced. At t_3, quality is observed and the two parties bargain. At t_4, final goods are produced and sold.

Interpretation: The timing creates the holdup problem: investments are made before quality and surplus division are contractible. Ex post bargaining then distorts ex ante investment incentives.

4.2 Incomplete contracts and ownership

The final-good producer and supplier cannot write enforceable contracts on input quality, ex ante investments, or sale revenues. They can contract only on ownership and transfers. Ex post surplus is divided by generalized Nash bargaining.

Under outsourcing, the supplier owns the intermediate input. Under integration, the final-good producer obtains residual control rights over a fraction of the input. Integration raises the final-good producer's outside option and lowers the supplier's bargaining share.

Interpretation: Integration does not eliminate holdup. It changes the distribution of bargaining power and therefore changes which party has stronger investment incentives.

4.3 Capital and labor investment asymmetry

Labor input is necessarily provided by the supplier. Capital expenditures are transferable: the final-good producer can rent capital and provide it to the supplier. This captures the idea that headquarters can control or finance capital more easily than labor effort.

Interpretation: This is the crucial asymmetry. If the input is capital intensive, giving the final-good producer stronger residual rights improves the incentive for the more important investment. If the input is labor intensive, outsourcing preserves supplier incentives.

4.4 Profits under integration and outsourcing

Let β_k be the capital intensity of the intermediate input in industry k. Define $\Theta(\beta_k)$ as the ratio of operating profits under vertical integration to those under outsourcing. The paper shows:

$$\Theta'(\beta_k) > 0.$$

That is, the relative attractiveness of integration rises with capital intensity.

Ownership form	**Main investment distortion**
Outsourcing	Supplier has stronger incentives; capital investment by the final-good producer is relatively weak.
Vertical integration	Final-good producer has stronger incentives; supplier's labor investment is relatively weak.

4.5 Proposition 1: industry capital intensity and integration

$$\exists\, \widehat{\beta} \in (0,1): \quad \begin{cases} \beta_k < \widehat{\beta} & \Rightarrow \text{outsourcing}, \\ \beta_k > \widehat{\beta} & \Rightarrow \text{vertical integration.} \end{cases}$$

Interpretation: This is the first key theoretical result. It predicts that capital-intensive goods are produced inside firm boundaries, while labor-intensive goods are outsourced.

4.6 General equilibrium and trade

The closed-economy model is embedded in a multicountry Helpman-Krugman setting. Countries differ only in factor endowments. Final goods are nontraded, while intermediate inputs are traded. Factor price equalization holds within a cone of diversification.

In this setting, the total imports of country N from country S satisfy

$$M^{N,S} = s^N s^S (rK + wL),$$

where s^N and s^S are country-size/spending shares.

The volume of intrafirm imports is

$$M_{i-f}^{N,S} = s^N s^S (rK + wL) \times \Psi\left(\frac{K^S}{L^S}\right),$$

where $\Psi(\cdot)$ is increasing in the exporting country's capital-labor ratio.

4.7 Lemma 1 and Proposition 2: country factor endowments

Lemma 1 states that the volume of country N's intrafirm imports from country S is increasing in the exporting country's capital-labor ratio, the exporting country's size, and the importing country's size.

Proposition 2 states that the *share* of intrafirm imports in total imports is increasing in the exporting country's capital-labor ratio and, conditional on capital abundance, is unaffected by relative country size:

$$\frac{\partial S_{i-f}^{N,S}}{\partial (K^S/L^S)} > 0.$$

Interpretation: The distinction between volume and share is important. Volumes depend on country size; shares isolate composition and depend on capital abundance.

5 Empirical design and identification

5.1 Industry-level test

The industry-level regression is the empirical analogue of Figure I:

$$\ln\left(S_{i-f}^{\text{US,ROW}}\right)_m = \theta_0 + \theta_2 \ln(K/L)_m + \Gamma' X_m + u_m.$$

The key prediction is

$$\theta_2 > 0.$$

Interpretation: The test asks whether more capital-intensive industries have higher intrafirm import shares, even after controlling for human capital, R&D intensity, advertising intensity, scale, and value added intensity.

5.2 Country-level share test

The country-level share regression is

$$\ln\left(S_{i-f}^{\mathrm{US},j}\right) = \gamma_0 + \gamma_1 \ln(K/L)_j + \gamma_2 \ln L_j + \Gamma' Z_j + u_j.$$

The theory predicts $\gamma_1 > 0$ and, for shares, no central role for country size.

5.3 Country-level volume test

The country-level volume regression is

$$\ln\left(M_{i-f}^{\mathrm{US},j}\right) = \omega_0 + \omega_1 \ln(K/L)_j + \omega_2 \ln L_j + \Gamma' Z_j + u_j.$$

The theory predicts both capital abundance and size should raise intrafirm import volumes:

$$\omega_1 > 0, \qquad \omega_2 > 0.$$

5.4 Identification logic

The empirical strategy is not a randomized design. It is a structural-consistency test of the model. The key identifying variation is cross-industry capital intensity and cross-country factor abundance. The main concern is that capital intensity or capital abundance may proxy for other variables that affect multinational integration. The paper addresses this by adding controls for human capital, R&D, advertising, scale, taxes, institutions, openness to FDI, and openness to trade.

Interpretation: The empirical design tests whether the comparative statics of the property-rights trade model survive obvious alternative explanations.

6 Tables and figures

6.1 Figure I: Share of Intrafirm U.S. Imports and Relative Factor Intensities; Figure II: Share of Intrafirm U.S. Imports and Relative Factor Endowments

Read: Figure I shows a positive cross-industry relationship between the intrafirm import share and industry capital intensity. Figure II shows a positive cross-country relationship between the intrafirm import share and exporter capital abundance.

Economic message: These two figures are the empirical puzzle that motivates the paper: intrafirm trade is systematically associated with capital, both at the industry level and at the country level.

The model must explain both slopes. Proposition 1 explains why capital-intensive industries are more integrated. Proposition 2 explains why capital-abundant countries export more through intrafirm channels.

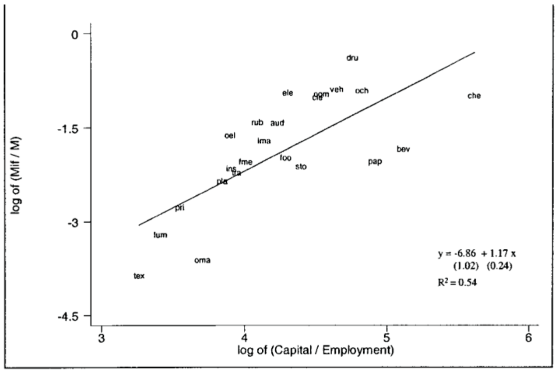


Figure I
Share of Intrafirm U. S. Imports and Relative Factor Intensities
The Y-axis corresponds to the logarithm of the share of intrafirm imports in

Figure I: intrafirm imports and industry capital intensity.

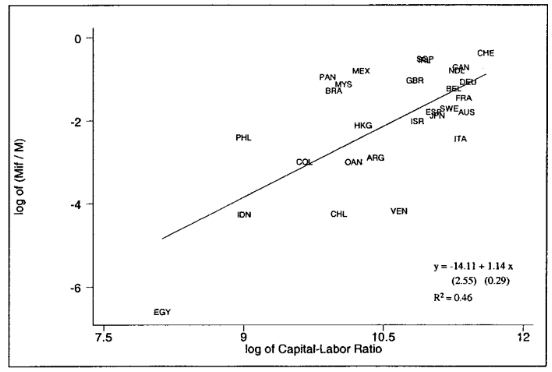


Figure II
Share of Intrafirm U. S. Imports and Relative Factor Endowments
The Y-axis corresponds to the logarithm of the share of intrafirm imports in

Figure II: intrafirm imports and country capital abundance.

The two motivating empirical correlations. Placing the figures on the same row highlights the parallel logic: capital intensity predicts integration by industry, and capital abundance predicts integration by country.

6.2 Table I: Decision-Making in U.S.-Based Multinationals; Figure III: Timing of Events

Read: Table I shows that U.S. multinational parents influence financial decisions more than employment/personnel decisions. Figure III lays out the model timing from ownership choice to investment, bargaining, and final production.

Economic message: Table I motivates the capital-versus-labor asymmetry; Figure III shows where this asymmetry enters the incomplete-contracting model.

Headquarters can more easily influence capital and financial choices than hiring and wage choices. This supports the assumption that capital investments are more transferable to the final-good producer than labor investments.

Decision-Making in U. S. Based Multinationals

% of British affiliates in which parent influence on decision is strong or decisive

Financial decisions		Employment/personnel decisions	
Setting of financial targets	51	Union recognition	4
Preparation of yearly budget	20	Collective bargaining	1
Acquisition of funds for working capital	44	Wage increases	8
Choice of capital investment projects	33	Numbers employed	13
Financing of investment projects	46	Layoffs/redundancies	10
Target rate of return on investment	68	Hiring of workers	10
Sale of fixed assets	30	Recruitment of executives	16
Dividend policy	82	Recruitment of senior managers	13
Royalty payments to parent company	82		

Source. Dunning [1993, p. 227]. Originally from Young, Hood, and Hamill [1985].

Table I: parent influence over financial and employment decisions.

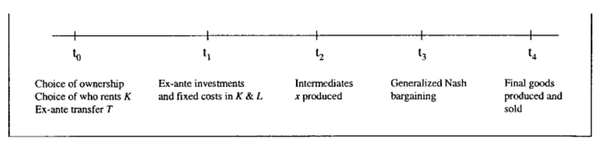


Figure III
Timing of Events

Figure III: timing of ownership, investment, bargaining, and production.

Organizational motivation and model timing. Table I supplies empirical motivation for the investment-transferability assumption; Figure III shows how this assumption generates holdup.

6.3 Figure IV: Complete Versus Incomplete Contracts; Figure V: Pattern of Production for $J = 2$

Read: Figure IV compares complete-contract investment choices with incomplete-contract equilibria under integration and outsourcing. Figure V shows the international allocation of intermediate production in a two-country economy.

Economic message: Figure IV is the property-rights mechanism. Figure V is the trade mechanism. Together they connect firm boundaries to the location of production.

Under integration, labor investment is relatively more distorted; under outsourcing, capital investment is relatively more distorted. Since capital-intensive industries prefer integration, capital-abundant countries specialize in integrated intermediate production.

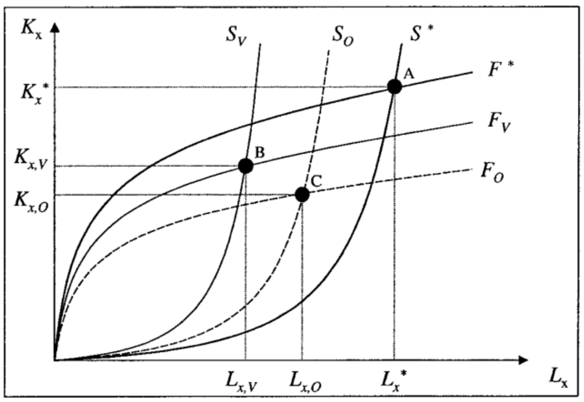


FIGURE IV

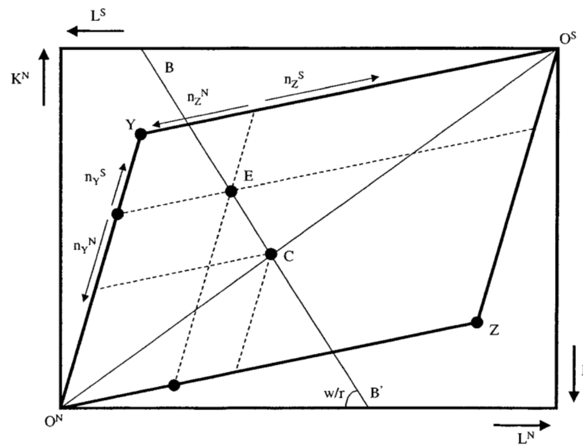


Figure IV: complete versus incomplete contracts.

Figure V: production pattern for two countries.

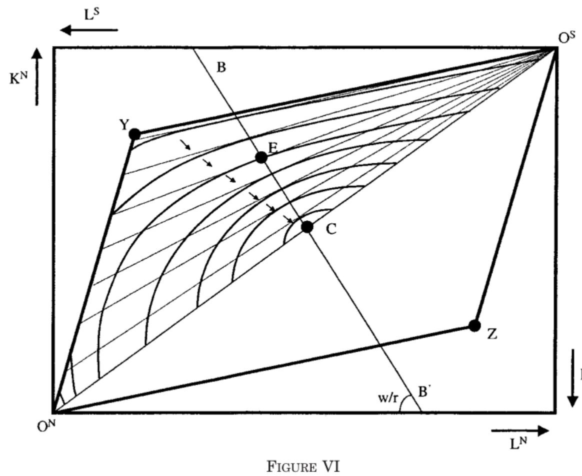
Mechanism pair: the first figure explains ownership incentives; the second embeds them in international specialization.

6.4 Figure VI: Volume of Intrafirm Imports; Figure VII: Share of Intrafirm Imports

Read: Figure VI shows iso-volume curves for intrafirm imports. Figure VII shows iso-share rays for the share of intrafirm imports.

Economic message: Volumes depend on country size and capital abundance; shares strip out country size and depend primarily on exporter capital abundance.

This visual distinction maps directly to Lemma 1 and Proposition 2. The country size terms matter for volumes but cancel out for shares.



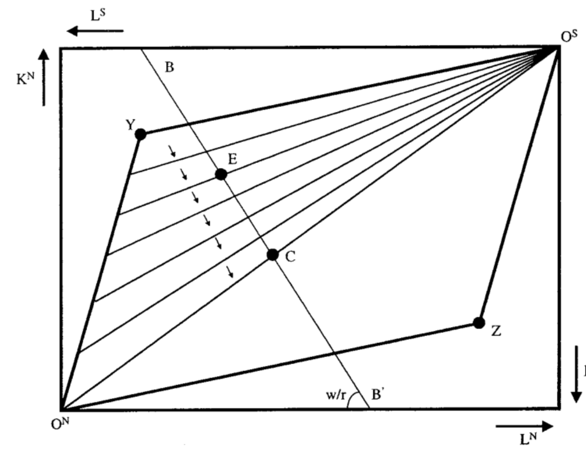


Figure VI: volume of intrafirm imports.   Figure VII: share of intrafirm imports.

Trade implication pair: Figure VI explains volumes; Figure VII explains shares.

6.5 Table II: Share of Intrafirm Imports in Total U.S. Imports; Table III: Descriptive Statistics

Read: Table II reports large variation in intrafirm import shares by industry and country. Table III summarizes all variables used in the regressions.

Economic message: Table II documents the raw variation to be explained; Table III shows that the econometric tests use substantial variation in capital intensity, size, human capital, R&D, and openness.

The range is economically large: industries such as drugs have high intrafirm shares, while textiles have low shares; capital-abundant countries such as Switzerland have much higher shares than capital-scarce countries such as Egypt.

SHARE OF INTRAFIRM IMPORTS IN TOTAL U. S. IMPORTS (PERCENT)

by Industry (avg. 1987–1994)				by Country (1992)			
DRU	65.5	FOO	13.9	CHE	64.1	ESP	15.5
OCH	40.9	PAP	12.7	SGP	55.4	AUS	15.5
VEH	39.8	FME	12.6	IRL	53.7	JPN	14.2
ELE	37.3	STO	11.8	CAN	45.1	ISR	12.4
COM	36.7	INS	11.1	NDL	42.2	HKG	11.2
CHE	35.9	TRA	10.7	MEX	41.7	PHL	8.4
CLE	35.7	PLA	9.1	PAN	35.8	ITA	8.1
RUB	23.9	PRI	6.1	GBR	33.2	ARG	5.1
AUD	23.8	LUM	4.1	DEU	31.9	COL	4.6
OEL	18.9	OMA	2.6	MYS	30.1	OAN	4.6
IMA	17.3	TEX	2.3	BEL	27.3	VEN	1.4
BEV	15.1			BRA	25.9	CHL	1.3
				FRA	21.6	IDN	1.3
				SWE	16.8	EGY	0.1

See Appendixes 4 and 5 for a list of industries and countries.

Table II: raw intrafirm import shares by industry and country.

TABLE III
DESCRIPTIVE STATISTICS

	Obs	Mean	St. dev.	Min	Max
ln $(S^{US,ROW}_{i-f})_m$	92	−1.90	0.92	−4.74	−0.19
ln $(K/L)_m$	92	4.26	0.57	3.21	5.73
ln $(H/L)_m$	92	−0.69	0.60	−1.78	0.60
ln $(R\&D/Sales)_m$	92	−4.20	1.00	−6.07	−2.47
ln $(ADV/Sales)_m$	92	−4.27	1.10	−6.63	−2.24
ln $(Scale)_m$	92	1.63	0.92	0.06	3.48
ln $(VAD/Sales)_m$	92	−0.66	0.18	−1.13	−0.32
ln $(S^{US,j}_{i-f})$	28	−2.08	1.44	−6.67	−0.45
ln $(K/L)_j$	28	10.54	0.86	8.13	11.59
ln $(L)_j$	28	16.03	1.20	13.63	18.16
ln $(H/L)_j$	28	0.82	0.19	0.47	1.10
$CorpTax_j$	28	0.32	0.08	0.15	0.44
$EconFreedom_j$	28	6.36	1.22	4.19	8.24
$OpFDI$	26	7.83	1.23	4.73	9.57
$OpTrade$	26	6.70	1.22	3.52	8.67
ln $(M^{US,j}_{i-f})$	28	6.36	2.64	−1.39	10.49

Table III: regression descriptive statistics.

Empirical foundation pair: the first table reports the dependent variable, and the second table reports the covariates.

6.6 Table IV: Factor Intensity and the Share of Intrafirm Imports; Table V: Factor Endowments and the Share of Intrafirm Imports

Read: Table IV shows that industry capital intensity predicts the intrafirm import share. Table V shows that exporter country capital abundance predicts the intrafirm import share.

Economic message: These are the main regression tests of the two key empirical predictions: capital-intensive industries and capital-abundant exporters have higher intrafirm trade shares.

In Table IV, the coefficient on industry capital intensity remains positive after controlling for human capital, R&D, advertising, scale, and value-added intensity. In Table V, the coefficient on exporter capital-labor ratio is positive even with size and institutional controls.

TABLE IV

FACTOR INTENSITY AND THE SHARE $S_{i-f}^{US,ROW}$

Dep. var. is ln $(S_{i-f}^{US,ROW})_m$	I	II	III	IV	V	VI
	Random effects regressions					
ln $(K/L)_m$	0.947***	0.861***	0.780***	0.776***	0.703***	0.723***
	(0.187)	(0.190)	(0.160)	(0.162)	(0.249)	(0.253)
ln $(H/L)_m$		0.369	−0.002	−0.038	−0.037	−0.081
		(0.213)	(0.188)	(0.200)	(0.206)	(0.221)
ln $(R\&D/Sales)_m$			0.451***	0.470***	0.452***	0.421***
			(0.107)	(0.114)	(0.128)	(0.140)
ln $(ADV/Sales)_m$				0.055	0.059	0.035
				(0.094)	(0.097)	(0.107)
ln $(Scale)_m$					0.068	0.100
					(0.179)	(0.190)
ln $(VAD/Sales)_m$						0.403
						(0.657)
R^2	0.50	0.55	0.72	0.73	0.73	0.73
No. of obs.	92	92	92	92	92	92
	Fixed effects regressions					
	I	II	III	IV	V	VI
ln $(K/L)_m$	0.599**	0.610**	0.610**	0.610**	0.943**	1.058**
	(0.299)	(0.300)	(0.300)	(0.300)	(0.412)	(0.410)
p-value Wu-Hausman test	0.14	0.27	0.62	0.64	0.52	0.19

Standard errors in parentheses (*, **, and, ***) are 10, 5, and 1 percent significance levels.

Table IV: industry capital intensity and intrafirm import shares.

FACTOR ENDOWMENTS AND THE SHARE $S_{i-f}^{US,j}$

Dep. var. is ln $(S_{i-f}^{US,j})$	I	II	III	IV	V	VI
ln $(K/L)_j$	1.141***	1.110***	1.244***	1.239***	1.097**	1.119**
	(0.289)	(0.299)	(0.427)	(0.415)	(0.501)	(0.399)
ln $(L)_j$		−0.133	−0.159	−0.158	−0.142	0.017
		(0.168)	(0.164)	(0.167)	(0.170)	(0.220)
ln $(H/L)_j$			−1.024	−0.890	−1.273	−0.822
			(1.647)	(1.491)	(1.367)	(1.389)
$CorpTax_j$				−0.601	0.068	1.856
				(3.158)	(3.823)	(2.932)
$EconFreedom_j$					0.214	
					(0.213)	
$OpFDI_j$						−0.384*
						(0.218)
$OpTrade_j$						0.292
						(0.273)
R^2	0.46	0.47	0.48	0.50	0.50	0.43
No. of obs.	28	28	28	28	28	26

Robust standard errors in parentheses (*, **, and, ***) are 10, 5, and 1 percent significance levels.

Table V: country capital abundance and intrafirm import shares.

Main evidence pair: Table IV tests Proposition 1's industry implication; Table V tests Proposition 2's country implication.

6.7 Table VI: Factor Endowments and the Volume of Intrafirm Imports

Read: Table VI studies volumes rather than shares. The exporter capital-labor ratio remains positive and significant, and exporter size also matters.

Economic message: This table validates Lemma 1. It shows that the volume of intrafirm imports rises with both exporter capital abundance and exporter size.

The volume result complements the share result. Shares are about composition; volumes are about both composition and scale. The theory predicts both and the data support both.

FACTOR ENDOWMENTS AND THE VOLUME $M_{i-f}^{US,j}$

Dep. var. is $\ln(M_{i-f}^{US,j})$	I	II	III	IV	V	VI
$\ln(K/L)_j$	2.048***	2.192***	2.188***	2.154***	1.650**	2.096***
	(0.480)	(0.458)	(0.716)	(0.663)	(0.762)	(0.695)
$\ln(L)_j$		0.607**	0.608**	0.614**	0.670**	0.700
		(0.229)	(0.268)	(0.271)	(0.243)	(0.419)
$\ln(H/L)_j$			0.031	0.953	−0.406	0.708
			(3.289)	(3.316)	(2.992)	(3.052)
$CorpTax_j$				−4.135	−1.763	−0.647
				(5.294)	(5.955)	(5.295)
$EconFreedom_j$					0.795	
					(0.443)	
$OpFDI_j$						−1.006**
						(0.474)
$OpTrade_j$						0.674
						(0.560)
R^2	0.44	0.52	0.52	0.53	0.60	0.49
No. of obs.	28	28	28	28	28	26

Robust standard errors in parentheses (*, **, and, ***) are 10, 5, and 1 percent significance levels.

Table VI: country factor endowments and intrafirm import volumes.

Result	Meaning
Prop. 1	Capital-intensive inputs are more likely to be integrated.
Lemma 1	Intrafirm import volumes rise with exporter size and capital abundance.
Prop. 2	Intrafirm import shares rise with exporter capital abundance.
Tables IV–VI	The model's comparative statics survive standard controls.

Notes-created result table: theoretical predictions and corresponding empirical tests.

Final empirical test and model-summary table. The left panel is the original Table VI; the right panel is a compact notes-created guide linking theory to evidence.

7 Short summary

- Intrafirm trade accounts for a large share of international trade.
- U.S. intrafirm import shares are higher in capital-intensive industries.
- U.S. intrafirm import shares are higher from capital-abundant countries.
- The model combines property-rights theory with monopolistic-competition trade.
- Incomplete contracts generate holdup in capital and labor investments.
- Vertical integration shifts residual rights toward the final-good producer.
- Capital-intensive industries integrate because capital is more transferable and controllable by headquarters.
- Labor-intensive industries outsource to preserve supplier labor incentives.
- Capital-abundant countries specialize in capital-intensive intermediates.
- Therefore they export a higher share of goods through intrafirm channels.
- The regression evidence supports both industry and country predictions.

8 Conclusion

8.1 Core conclusion

Antràs shows that firm boundaries are central to understanding the structure of international trade. The pattern of intrafirm trade is not random. It is systematically related to industry capital intensity and country factor endowments because ownership changes the incentives to make relationship-specific investments.

8.2 Economic intuition

Incomplete contracts create holdup. Integration gives the final-good producer more control rights and stronger incentives to invest in capital, while outsourcing gives the supplier stronger incentives to invest in labor. Therefore capital-intensive production is more likely to be vertically integrated, while labor-intensive production is more likely to be outsourced.

8.3 Why the trade model matters

The property-rights model explains why capital-intensive sectors are integrated. The trade model explains why capital-abundant countries have high intrafirm trade shares: they specialize in those capital-intensive sectors. This is the key link between firm theory and comparative advantage.

8.4 Incremental contribution revisited

The paper's contribution is to make the boundary of the firm an endogenous object in a general-equilibrium trade model. It moves beyond models in which multinationals exist only because of transport costs or scale economies, and beyond property-rights models that abstract from international trade.

8.5 Limitations

The model assumes Cobb-Douglas production, factor price equalization, identical technologies across countries, and a specific asymmetry between capital and labor contractibility. The empirical tests use U.S. related-party import data and broad industry/country proxies. The results are therefore best read as structural consistency evidence, not as randomized causal estimates.

8.6 Takeaway

The paper's lasting message is that the organization of firms and the pattern of trade must be studied together. Comparative advantage determines what countries produce, and incomplete contracts determine whether that production is organized inside multinational firms or through arm's-length suppliers.